WHEN DOES CELL CONDITIONING HARM RECONSTRUCTION? DIAGNOSTICS FOR ARBITRARY-SCALE SUPER-RESOLUTION

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ABSTRACT

Relative to zero cell, native target-cell conditioning in arbitrary-scale super-resolution improves average quality but need not improve every image or frequency band. We diagnose this discrepancy through fixed-checkpoint corrections, separating amplitude overshoot from wrong-direction changes using an exact squared-error identity. A prospective replication on 32 natural images, two model families and two training seeds finds harmful corrections against both zero and nonzero cell references. Relative to zero cell, native conditioning improves mean PSNR in all four model–scale groups, yet uniform half-correction increases mean RGB MSE by 0.26–7.55%. Nonzero references reveal additional failures and change which images are labelled harmful: failure is a property of a specified correction, not an intrinsic image label. Controlled high-frequency input paths and equal-power phase changes further produce opposite gains at fixed scale. Together, these measurements explain why average frequency ordering, local band damage and successful rescue of selected cases are insufficient criteria for a globally beneficial readout correction.

***Index Terms*—** Arbitrary-scale super-resolution, cell conditioning, frequency recovery, readout failure analysis

1. INTRODUCTION AND RELATED WORK

Arbitrary-scale super-resolution (ASISR) predicts image values at continuous coordinates. LIIF supplies target cell size to a local decoder [1]; LTE introduces cell-dependent phase estimation [2]. This information specifies the requested scale but does not guarantee that the resulting output change reduces reconstruction error. Our question is: *at the same scale and frequency band, why can a cell-conditioned correction help one image and hurt another?* We examine output-level failure types and controlled conditions that change benefit, without identifying an internal neural mechanism.

Harmful cell effects are not new. LIIF already varies cell size at fixed queries; IPE-LIIF integrates positional features and reports cases where an additional cell input hurts [3]. Thera supplies analytic Gaussian filtering and discusses its mismatch with bicubic training [4]. Our fixed-checkpoint interventions are not a performance comparison against these methods. Integrating intermediate features also need not equal integrating a nonlinear decoder's RGB output.

Frequency-resolved analysis has precedents in HyRA's linear/nonlinear responses [5], texture-based error analysis [6], sinusoidal degradation probes [7], and Nyquist-normalised forecasting [8]. The importance of phase beyond spectral magnitude is classical [9]. We do not claim these principles, or the quadratic identity below, as new. A wider continuous-SR benchmark studies architecture and retraining effects [10]; we isolate interventions on inherited models.

We study cell conditioning as an output correction with measurable benefits and costs. The empirical contributions are: (1) paired natural-image measurements that separate image-level from band-level harm; (2) an overshoot/direction diagnosis, using a classical error decomposition, with an explicit test of the cost of uniform attenuation; and (3) controlled input-path and fixed-power phase experiments that expose conditions hidden by coarse frequency summaries. The argument follows observed failures, their error geometry, and interventions that change their sign.

2. EVALUATION OBJECTS AND FREQUENCY RECOVERY

2.1. Keep the comparison fixed

Let $y = Ax$ denote the LR observation, $Q_h x$ the desired target measurement, and c the cell supplied to $F_\theta(y, q, c)$. Changes to A, Q_h and c answer different questions. Natural-image tests use discrete HR targets and bicubic antialiased downsampling, not certified box averages of a known continuous field. Assigning $c = h$ does not establish that the model implements the physical operator Q_h.

The LIIF/LTE/SC-INR variants inherit an EDSR-baseline encoder, DIV2K training [11], 1000 epochs and training scales $\times 1$–$\times 4$. SC-INR is the examined sinc-modulated Fourier-feature implementation. A trained full/no-readout ablation changes weights; a native/zero-cell comparison holds weights, observation and queries fixed. Zero cell is outside the training cell support. Earlier probes use one seed1 checkpoint per LTE/SC model; the new natural-image replication uses both inherited seeds 1 and 2. The LTE snapshots omit the original paper's extrapolation cell clamp. All interventions preserve

weights.

2.2. A useful coordinate with image-wise exceptions

For image i, band b, target energy $S_{ib} = \|\Pi_b(x_i - \bar{x}_i)\|^2$ and error energy $E_{ib} = \|\Pi_b(x_i - \hat{x}_i)\|^2$, define

$$R_b = 1 - \frac{\sum_i E_{ib}}{\sum_i S_{ib}}, \qquad D = \mathrm{median}_b[\max_s R_b(s) - \min_s R_b(s)]. \tag{1}$$

The frequency study uses the RGB mean, unwindowed FFTs, centre HR240 crops and respectively 14, 30 and 30 images from Set14, BSD100 and Urban100 [12, 13, 14]. With Chebyshev frequency $\nu = \max(|\nu_x|, |\nu_y|)$ and scale $s = 2, 3, 4$, seven bins span $.25, .5, \ldots, 2$ in $u = 2s\nu$; $u = 1$ is LR Nyquist. The absolute control fixes bin edges at $\nu = u/6$. Different bin supports compare evaluation coordinates, not equality of reconstructed functions.

Normalisation reduces D in all 27 model–dataset–seed groups; primary values fall from .31–.46 to .013–.061. This is a change in evaluation coordinates, not a reconstruction improvement. All 24 saved-output sensitivity settings retain the reduction. However, the original .05 tolerance passes only 19/27 pooled groups, versus 27/27 equal-image groups. The repetitions share images and are not independent experiments.

Pooling can conceal the relevant variation. In Urban100, all nine pooled curves pass .05, while only 14–17 of 30 images per group pass the same descriptive tolerance. Holding image weights fixed across scales increases D in all nine Urban100 groups, although it decreases D in all nine Set14 groups. These findings motivate image-paired diagnosis: an aggregate frequency curve cannot decide whether an individual image benefits.

3. FROM FREQUENCY ORDERING TO FAILURE TYPES

3.1. Coarse ordering is an insufficient explanation

In LTE/SC seeds 2 and 3, high-band full/no-readout ablation loss exceeds middle-band loss in 34/36 dataset–scale settings. Yet the tested footprint score has exactly the same ranks as band index. A separate fixed-query conditioning-average test preserves ordering in 4/4 groups, but matches its function-similarity criterion in 0/4. Ordering alone therefore does not establish an integral readout interpretation. Retrained ablations and fixed-weight interventions also answer different questions: on barbara at $\times 2$, SC's high-band recovery gain is $-.193296$ for the trained pair but $+.009733$ for native/zero cell. We therefore hold the model fixed in the following diagnosis.

3.2. Overshoot and wrong direction

For fixed-weight outputs, let F_c be native-cell prediction, F_r a declared reference-cell prediction, t the HR target, and $d = F_c - F_r$. Inner products are pixel/channel means, optionally after band projection. Set $a = \langle t - F_r, d\rangle$, $b = \|d\|^2$ and

$$G(\lambda) = \mathrm{MSE}(F_r, t) - \mathrm{MSE}(F_r + \lambda d, t) = 2\lambda a - \lambda^2 b. \tag{2}$$

For a harmful native correction $G(1) < 0$ and $b > 0$, $0 < a < b/2$ is *overshoot*: a smaller positive correction can help. When $a < 0$, the *direction is wrong*: every positive multiple hurts. The oracle optimum is $\lambda^* = a/b$. This classical identity diagnoses saved outputs using HR truth; output interpolation is not a cell intervention or an observation-only method. We define harm by $G(1) < -10^{-8}$. For the saved natural outputs, b, $|a|$ and $|a - b/2|$ within $64\epsilon_{64}\max(\mathrm{MSE}(F_r, t), b, 10^{-8})$ are retained as numerical boundaries; ϵ_{64} is machine precision.

3.3. Development diagnosis and prospective replication

An initial saved-output diagnosis uses 32 development images (DIV2K train0001–0032), centre HR128, $\times 2/4$, shave4 and raw RGB. Against zero cell, LTE has 13 harmful bands (12 overshoot, 1 wrong direction), and SC has 27 (14, 13). Halving the output correction rescues 11/13 and 6/27 bands, yet increases overall RGB MSE by 2.405% and .966%. This motivates replication of both the failures and their aggregate cost, rather than treating a rescued subset as a successful policy.

We freeze a new cohort, DIV2K valid0837–0868, before inference: 32 images, LTE/SC seeds 1 and 2, and $\times 2/4$. These images are separate from the previous development and forecast evaluation cohorts and the checkpoints' first-ten-image validation subset. The centre HR128 target and bicubic-AA LR observation are held fixed. Each checkpoint predicts four supplied relative cells $c_{\mathrm{rel}} = c\, n_{\mathrm{LR}} \in \{0, 1/2, 2/3, 1\}$; native is 1 at $\times 2$ and $1/2$ at $\times 4$. Training uses LR48 and integer HR48–192, so all three nonzero values occur in the training cell's numeric support. Changing cell at a fixed target remains mismatched conditioning, not a joint in-distribution sample or physical target change.

All 1024 RGB predictions are hash-frozen before target scoring. The primary contrasts are native versus zero and native versus $c_{\mathrm{rel}} = 2/3$ (denoted $h3$); the other nonnative contrasts are retained. Three non-DC Chebyshev bands use cutoffs $1/8, 1/4, 1/2$ on the shaved RGB channels, with pixel-domain normalisation. Image-cluster bootstrap resamples all 32 images jointly across seeds, 10,000 times, with pointwise intervals; repeated seeds and scales are not additional independent images.

Average benefit coexists with individual harm. Against zero, mean native-minus-reference PSNR is $.912/.444$ dB for LTE at $\times 2/4$ and $.687/.028$ dB for SC; all four 95% image-bootstrap intervals are above zero. Yet Table 1 contains negative cases in every group. The nonzero $h3$ contrast also has

Table 1. Prospective natural-image replication. Each row has 64 RGB and 192 band records (32 images, two seeds). RGB: all negative records followed by those losing $> .01$ dB in parentheses. Band O/D: overshoot/ wrong-direction counts. $\Delta E_{1/2}$: half-correction mean RGB MSE change relative to native, over all cases.

Model	s	Ref.	RGB	Band O/D	$\Delta E_{1/2}$
LTE	2	zero	1 (1)	5/1	+7.55%
	2	$h3$	13 (6)	13/11	+2.29%
	4	zero	2 (1)	18/2	+2.77%
	4	$h3$	35 (10)	21/84	$-.058\%$
SC	2	zero	10 (5)	10/12	+5.46%
	2	$h3$	17 (9)	13/16	+1.84%
	4	zero	11 (5)	17/30	+.264%
	4	$h3$	25 (7)	11/83	+.087%

harmful images in both seeds of each model and scale, so harm is not confined to zero-cell probes. At $\times 4$, 35/64 LTE and 25/64 SC RGB records are negative, but only 10 and 7 lose more than .01 dB; worst losses are .202 and .136 dB. Counts and effect sizes answer different questions.

Failure identity depends on the correction. SC at $\times 2$ retains all ten zero-reference negative RGB records against $h3$. At $\times 4$, its eleven zero-reference failures and 25 $h3$ failures are disjoint. Here the references lie on opposite sides of native cell. This is a contrast change, not evidence of a new internal mechanism. Across the two training seeds, zero-reference SC RGB harm repeats in 10/11 image–scale cases harmed by either seed; LTE repeats in 1/2. Against $h3$, the corresponding overlaps are 17/25 and 19/29. Failure types occur repeatedly, but exact labels and their prevalence cannot be assigned permanently to a model or image.

Local rescue does not imply a useful global policy. For the zero contrast, half-correction raises aggregate RGB MSE in all four model–scale groups (Table 1), with all four 95% intervals above zero. The same holds for $h3$ at $\times 2$; at $\times 4$ both intervals include zero. Thus the supported result is a repeatable cost of this attenuation in the specified contrasts, not failure of every possible global weight. Band losses must also be separated from RGB losses: improvements in other bands and DC can offset a damaged band.

4. CONTROLLED INPUT-PATH AND PHASE EXPERIMENTS

4.1. Design: separate two interventions

To probe conditions behind alignment changes, we keep the two checkpoints and $\times 4$ scale fixed. A monochrome HR64 target, replicated across RGB, is $t = .5 + L + H$, with three low cosine modes $L : \{(2,0),(0,2),(2,2)\}$ with zero initial phases. All six cosine modes have amplitude .04. Frequencies are cycles per 64 pixels. Four H templates are

$$M_a = \{(10,0),(12,0),(14,0)\},$$
$$M_c = \{(10,0),(0,12),(10,10)\},$$
$$H_a = \{(18,0),(22,0),(26,0)\},$$
$$H_c = \{(18,0),(0,22),(18,18)\}.$$

All lie above LR Nyquist (8 cycles). A low-frequency sham uses $\{(4,0),(0,4),(4,4)\}$. Four high-branch phase vectors (one zero and three uniform vectors, seed 20261012) are crossed with four translations $(0,0),(.5,0),(0,.5),(.5,.5)$ in LR-pixel units. Translations move both branches and are nested position controls, not independent mechanisms.

For each fixed target, a periodic separable bicubic-AA operator A produces LR16 observations

$$y_\eta = A(.5 + L) + \eta A(H), \qquad \eta \in \{0, .5, 1\}. \tag{3}$$

The sham always uses At. Thus η controls a specified high-frequency input path; it also changes LR energy/gradients, not aliasing independently of input statistics. Separately, changing high-branch phase holds each Fourier magnitude fixed while changing the target. All modes are even, so both full64 and primary centre32 regions contain integer periods. No observations or outputs are clipped. Bands use the same three cutoffs as the natural-image diagnosis, with DC excluded; gain ties use 10^{-8}, alignment boundaries 10^{-10} and tiny-change energy 10^{-12}. The 240 conditions contain 148 unique inputs and 296 native/zero prediction pairs, not 240 independent images. Inputs and then all predictions were hash-frozen before scoring; this is a deterministic diagnostic, not a blinded natural-image test.

4.2. Alignment reversals and fixed-power counterexamples

For the two middle-frequency templates (32 phase–position conditions), increasing η from 0 to 1 lowers middle-band gain in 29/32 LTE and 32/32 SC conditions. Harmful outcomes increase from 18 to 32 and 15 to 31, respectively; 14/32 and 16/32 change from positive to negative alignment a. Full64 evaluation retains gain decreases in 31/32 for both models, although exact negative counts change. The high-frequency templates instead contain both signs of path effect across the fixed grid; the path is not uniformly harmful.

At $\eta = 1$, changing phase at fixed template and position produces both positive and negative high-band gains in all eight template–position groups per model, in both evaluation regions. Thus *target power alone does not determine readout benefit in these fixed models*. This does not hold LR power/gradients fixed. At $\eta = 0$, high-phase variants share identical inputs and outputs: score differences then reflect different unobserved targets, not a neural response to hidden phase. These distinctions prevent phase sensitivity from being mistaken for a general adaptive mechanism.

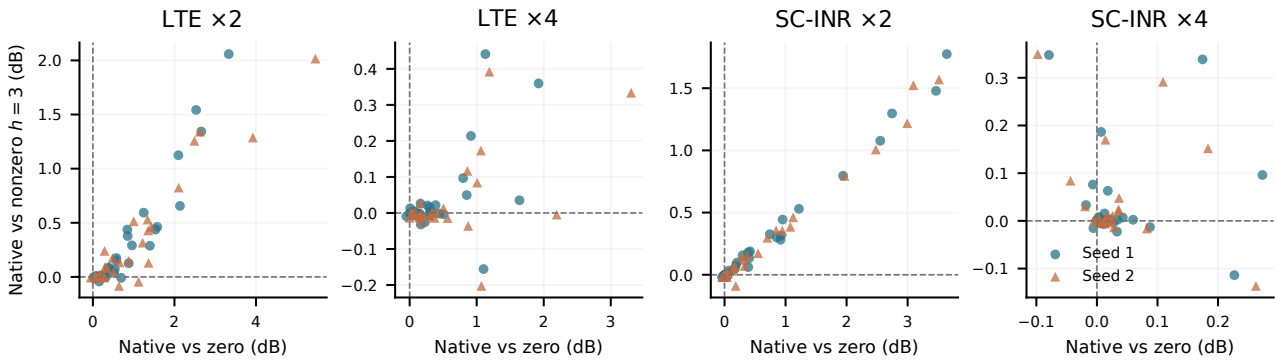


Fig. 1. All 32 natural images under both seeds, paired at fixed model and scale. Coordinates are full-RGB native-minus-reference PSNR gains. Each point compares the same image under zero and nonzero $h3$ references; negative coordinates indicate harm. Reference choice changes the harmful subset, particularly when it changes the direction of the cell intervention.

Effects are small: at $\eta = 1$ in the high templates, centre-region full-RGB native-minus-zero PSNR spans $[-.00730, .01236]$ dB for LTE and $[-.00164, .000891]$ dB for SC. Sign reversals provide diagnostic counterexamples, not a practical performance advance or an estimate of natural-image causal prevalence.

5. GUIDANCE, SCOPE AND CONCLUSION

Explanation does not yet supply a selector. For bicubic upsampling $B(y)$, the observable proxy replaces t in Eq. (2); true gain minus proxy gain is exactly $2\langle t - B(y), d\rangle$. Unlike generalized SURE under its stated assumptions [15], this proxy has no unbiased-risk guarantee. With 32 calibration and 12 separate evaluation images (valid0825–0836), forecasts were frozen before scoring. SC's magnitude/spectrum ridge detects 6/14 harmful bands with 15 false positives; adding proxy alignment still detects six with 17 false positives (balanced accuracy .585 versus .568). Neither candidate establishes a reliable RGB policy gain; LTE has only one harmful band outcome. Saved-output crop/band sensitivity retains SC harm without repairing the forecasts.

A second, retrospective test uses lower-resolution reconstruction gains with eight-fold whole-image-held-out fitting on the 32 development images. Both paired image-bootstrap AUC improvements include zero. Directly selecting native/zero by the lower-level gain increases RGB MSE by .239% (LTE) and .142% (SC). Thus neither tested proxy provides a reliable operational substitute for the HR-assisted diagnosis.

Scope and cost. The strongest natural-image evidence concerns the declared output contrasts on two model families and two training seeds. Synthetic periodicity and a finite phase grid limit the controlled examples; they do not identify the causal prevalence of aliasing in natural images. The zero reference is unsupported, while the nonzero controls establish numeric cell support only. No matching training or new architecture is introduced. Independent pixel/FFT and bootstrap recomputation verifies saved scoring, rather than reimplementing the networks. The new complete experiment takes 637.27 seconds on one CPU thread, with 6.88 seconds for independent audit and no GPU; excluded-image preflight inference/scoring takes 27.28 seconds, followed by a .17-second scoring repair. The original failed serialization is retained. The earlier controlled experiment takes 85.32 seconds on one CPU thread.

Conclusion. Relative to zero cell, native conditioning improves average reconstruction in the tested models, while individual corrections can overshoot or move away from the target. The new natural-image replication establishes failures against nonzero references and across training seeds, while showing that their identities remain contrast-dependent. Controlled input paths and fixed-power phase changes expose structure omitted by band-energy summaries. A useful assessment of readout must therefore report the specified correction, its failure type, the actual image-level loss, and the aggregate cost of any proposed rescue. The present evidence supports this diagnostic framework; a reliable observation-only correction rule remains open.

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